

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1758

Roanoke Rapids 138 kV
July 04, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1758. The Interconnection Customer proposes a 71.5 MW battery energy storage facility interconnecting at the Roanoke Rapids 138 kV substation in the North Carolina area. The request was received on May 23, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 601 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 23 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,873,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: N Tower Tap / Cedar Falls Wi 138 kV
Contingency: Haggin / N Tower Tap 345 kV L
Loading: 134.4% (Rating: 304 MVA)
DFAX: 6.4%
Pre-project loading: 132.9%
2. Facility: Seco Substation / Wing Lane 138 kV
Contingency: Mitchell Dam / Skaggs Island Junction 2 345 kV
Loading: 119.9% (Rating: 413 MVA)
DFAX: 9.9%
Pre-project loading: 118.2%
3. Facility: Spring Street / Wing Lane 138 kV
Contingency: Mitchell Dam Substation / Skaggs Island Junction 2 345 kV
Loading: 119.2% (Rating: 309 MVA)
DFAX: 7.5%
Pre-project loading: 117.5%
4. Facility: Harvard / Seco 138 kV Line
Contingency: Mitchell Dam / Skaggs Island Jct 2 345 kV
Loading: 116.3% (Rating: 524 MVA)
DFAX: 9.9%
Pre-project loading: 114.9%
5. Facility: Redbed Plains Wind Farm / Fruitridge & S. Land Park 138 kV L
Contingency: Mitchell Dam / Skaggs Island Jct 2 345 kV
Loading: 112.4% (Rating: 341 MVA)
DFAX: 6.2%
Pre-project loading: 111.1%

6. Facility: N Tower Tap / Dpc Elroy 138 kV L
Contingency: Mccartney / N Tower Tap 345 kV
Loading: 112.3% (Rating: 661 MVA)
DFAX: 6.2%
Pre-project loading: 111.6%
7. Facility: Mickens / Igo 345 kV
Contingency: Mickens / Newport News 345 kV
Loading: 111.6% (Rating: 1,089 MVA)
DFAX: 8.8%
Pre-project loading: 111.1%
8. Facility: Carlin / Fruitridge & S. Land Park 138 kV
Contingency: Mitchell Dam / Skaggs Island Jct 2 345 kV
Loading: 111.2% (Rating: 385 MVA)
DFAX: 6.2%
Pre-project loading: 110.1%
9. Facility: Birchwood / Dist 1000 Substation 138 kV
Contingency: Birchwood / Leavitt 345 kV
Loading: 110.0% (Rating: 619 MVA)
DFAX: 5.1%
Pre-project loading: 109.4%
10. Facility: Trentn Diesel / Dist 1000 138 kV
Contingency: Leavitt / Birchwood 345 kV
Loading: 109.7% (Rating: 633 MVA)
DFAX: 5.1%
Pre-project loading: 109.1%
11. Facility: Haggin 345 kV/138 kV Auto
Contingency: N Tower Tap Substation / Haggin 345 kV
Loading: 109.5% (Rating: 426 MVA)
DFAX: 5.3%
Pre-project loading: 108.6%
12. Facility: Spring St / Chipola 138 kV
Contingency: Mitchell Dam / Skaggs Island Junction 2 345 kV
Loading: 108.7% (Rating: 300 MVA)
DFAX: 7.5%
Pre-project loading: 107.0%
13. Facility: Dunn Siding / Birchwood 345 kV
Contingency: Leavitt Substation / Birchwood 345 kV L
Loading: 107.4% (Rating: 1,519 MVA)
DFAX: 7.5%
Pre-project loading: 107.0%
14. Facility: Dunn Siding / Fredericksburg 345 kV
Contingency: Nauvoo / Birchwood 345 kV
Loading: 106.6% (Rating: 1,914 MVA)
DFAX: 14.9%
Pre-project loading: 106.0%
15. Facility: Redbed Plains Wind Farm / Westcoast Cellufibre Substation 138 kV
Contingency: Mitchell Dam / Skaggs Island Junction 2 345 kV
Loading: 106.3% (Rating: 528 MVA)
DFAX: 6.2%
Pre-project loading: 105.4%
16. Facility: Kings River / Dg Amp Solar Coldwater 138 kV L
Contingency: Skaggs Island Jct 2 / Mitchell Dam 345 kV Line

Loading: 104.7% (Rating: 435 MVA)

DFAX: 5.1%

Pre-project loading: 103.9%

17. Facility: Mickens / Newport News 345 kV Line

Contingency: Mickens / Igo 345 kV

Loading: 103.7% (Rating: 1,268 MVA)

DFAX: 10.1%

Pre-project loading: 103.2%

18. Facility: Westcoast Cellufibre / Leavitt 138 kV Ckt 1

Contingency: Mitchell Dam / Skaggs Island Junction 2 345 kV

Loading: 102.9% (Rating: 469 MVA)

DFAX: 8.1%

Pre-project loading: 101.6%

19. Facility: Westcoast Cellufibre / Leavitt 138 kV

Contingency: Mitchell Dam / Skaggs Island Jct 2 345 kV line

Loading: 102.9% (Rating: 469 MVA)

DFAX: 8.1%

Pre-project loading: 101.6%

20. Facility: McCartney / Gaston 345 kV

Contingency: McCartney / Roanoke Rapids 345 kV Line

Loading: 102.1% (Rating: 1,659 MVA)

DFAX: 18.0%

Pre-project loading: 101.4%

21. Facility: Chancellor Row / Bear Island 345 kV

Contingency: Leavitt / Birchwood 345 kV

Loading: 102.1% (Rating: 2,686 MVA)

DFAX: 24.1%

Pre-project loading: 101.5%

22. Facility: North Anna / Dunn Siding 345 kV

Contingency: Fredericksburg / Dunn Siding 345 kV

Loading: 101.8% (Rating: 1,109 MVA)

DFAX: 13.8%

Pre-project loading: 100.9%

23. Facility: N Tower Tap / Haggin 345 kV

Contingency: Bear / Mecklenburg Power Sta 345 kV

Loading: 101.6% (Rating: 1,490 MVA)

DFAX: 30.1%

Pre-project loading: 100.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
N Tower Tap / Cedar Falls Wi 138 kV	Rebuild line with higher-capacity conductor	\$8,840,000	4% (\$316,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Seco Substation / Wing Lane 138 kV	Rebuild line with higher-capacity conductor	\$1,886,000	1% (\$19,000)
Sprin Street / Wing Lane 138 kV	Rebuild line with higher-capacity conductor	\$1,895,000	1% (\$24,000)
Harvard / Seco 138 kV Line	Rebuild line with higher-capacity conductor	\$4,275,000	1% (\$42,000)
Redbed Plains Wind Farm / Fruitridge & S. Land Park 138 kV L	Rebuild line with higher-capacity conductor	\$14,949,000	1% (\$110,000)
Dpc Elroy / North Tower Tap Substation 138 kV	Rebuild line with higher-capacity conductor	\$5,147,000	1% (\$72,000)
Mickens / Igo 345 kV line	Rebuild line with higher-capacity conductor	\$46,664,000	1% (\$533,000)
Carlin / Fruitridge & S. Land Park 138 kV	Rebuild line with higher-capacity conductor	\$2,181,000	1% (\$16,000)
Dist 1000 / Birchwood 138 kV line	Rebuild line with higher-capacity conductor	\$31,407,000	1% (\$377,000)
Trentn Diesel / Dist 1000 138 kV	Reconductor line	\$12,813,000	1% (\$154,000)
Haggin 345/138 kV TX	Replace transformer with higher-rated unit	\$26,133,000	1% (\$214,000)
Spring St / Chipola 138 kV	Reconductor line	\$3,337,000	1% (\$42,000)
Dunn Siding / Birchwood 345 kV	Reconductor line	\$21,820,000	1% (\$124,000)
Fredericksburg / Dunn Siding 345 kV	Reconductor line	\$17,441,000	1% (\$100,000)
Redbed Plains Wind Farm / Westcoast Cellufibre Substation 138 kV	Reconductor line	\$1,158,000	1% (\$7,000)
Kings River / Dg Amp Solar Coldwater 138 kV L	Reconductor line	\$789,000	1% (\$5,000)
Newport News / Mickens 345 kV	Reconductor line	\$11,548,000	1% (\$121,000)
Westcoast Cellufibre / Leavitt 138 kV Ckt 1	Reconductor line	\$681,000	1% (\$5,000)
Westcoast Cellufibre / Leavitt 138 kV Ckt 2	Reconductor line	\$724,000	1% (\$5,000)
Mccartney / Gaston 345 kV	Reconductor line	\$25,461,000	1% (\$248,000)
Chancellor Row / Bear Island 345 kV	Reconductor line	\$8,864,000	1% (\$53,000)
North Anna / Dunn Siding 345 kV line	Reconductor line	\$15,921,000	1% (\$96,000)
N Tower Tap / Haggin 345 kV line	Reconductor line	\$14,251,000	1% (\$190,000)
	Total allocated to project		\$2,873,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.